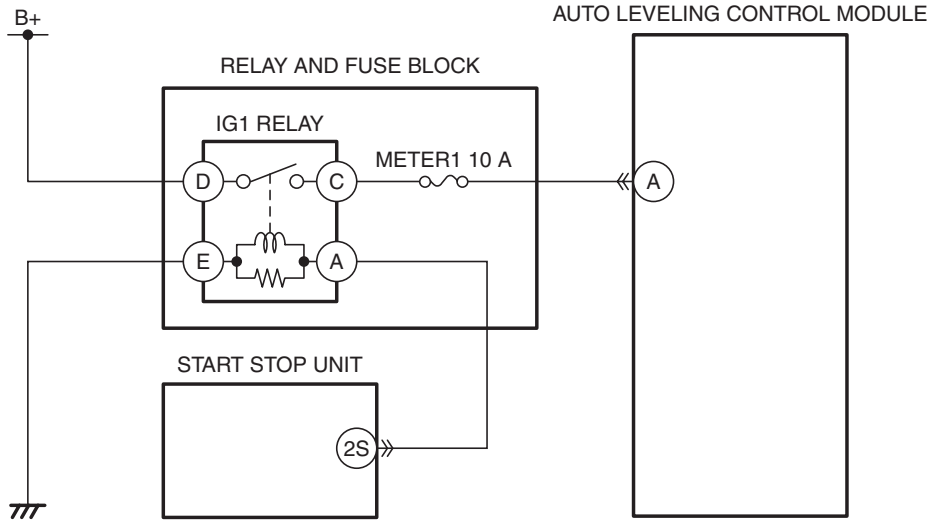
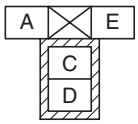
DTC U3003:16 [HEADLIGHT AUTO LEVELING SYSTEM]

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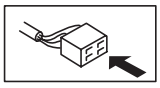
Description	Auto leveling control module low power supply voltage input
Detection condition	• Auto leveling control module power supply circuit voltage of **9.0 V or less** is detected for **5 s or more** with the ignition switched ON (engine off or on).
Fail-safe	• The headlight leveling actuator is stopped at the position when the malfunction is determined.
Possible cause	• DTCs are stored in the PCM. • Battery malfunction • Generator malfunction • Auto leveling control module connector or terminal malfunction • Auto leveling control module power supply circuit malfunction — Short to ground in wiring harness between METER1 10 A fuse and auto leveling control module terminal A — METER1 10 A fuse malfunction — Open circuit in wiring harness between IG1 relay terminal C and auto leveling control module terminal A • Auto leveling control module malfunction



IG1 RELAY
(RELAY AND FUSE BLOCK)

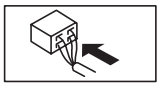


FRONT



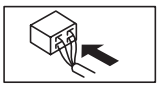
AUTO LEVELING CONTROL MODULE
WIRING HARNESS-SIDE CONNECTOR

AC	AA	Y	W							I	G	E	C	A
AD	AB	Z	X	V	T	R	P	N	L	J	H	F	D	B



START STOP UNIT
WIRING HARNESS-SIDE CONNECTOR

2W	2U	2S	2Q	2O	2M	2K	2I	2G	2E	2C	2A
2X	2V	2T	2R	2P	2N	2L	2J	2H	2F	2D	2B



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY PCM DTCs** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is the battery normal?	Yes	Go to the next step.
		No	Recharge or replace the battery, then go to Step 6. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3	**INSPECT GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-D 1.5)].) • Is the generator normal?	Yes	Go to the next step.
		No	Replace the generator, then go to Step 6. (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
4	**INSPECT AUTO LEVELING CONTROL MODULE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the auto leveling control module connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 6.

Step	Inspection		Action
5	**VERIFY AUTO LEVELING CONTROL MODULE POWER SUPPLY VOLTAGE** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Display PID VPWR_IG using the M-MDS. (See PID/DATA MONITOR INSPECTION [HEADLIGHT AUTO LEVELING SYSTEM].) • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the METER1 10 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between METER1 10 A fuse and auto leveling control module terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between IG1 relay terminal C and auto leveling control module terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
6	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the auto leveling control module using the M-MDS. (See CLEARING DTC [HEADLIGHT AUTO LEVELING SYSTEM].) • Switch the ignition ON (engine off or on) and wait for **5 s or more**. • Retrieve the auto leveling control module DTCs using the M-MDS. (See DTC INSPECTION [HEADLIGHT AUTO LEVELING SYSTEM].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the auto leveling control module. (See AUTO LEVELING CONTROL MODULE REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
7	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [HEADLIGHT AUTO LEVELING SYSTEM].)
		No	DTC troubleshooting completed.